

was to become brittle and break.

3D printing metals is also gaining popularity over the past couple of years. Gold and silver have been added to the range of metal materials that can be 3D-printed directly, with applications across the jewellery sector. These are both very strong materials and are processed in powder form. With less than 0.1 per cent residual ash after burnout, Wax Cast from Makerjuice Labs is a relatively-new resin being used by jewellery makers. Optimised for 25µm to 50µm layers, it is resistant to most solvents including alcohol and acetone.

Another material being tested out is aluminium. 3D-printed aluminium is known to be great for producing strong, light and precise end-use metal parts. Post printing, the parts look more matte and feature rougher surfaces than traditionally-milled aluminium. Applications of such materials include functional

parts, spare parts and jewellery.

Ceramics is a relatively newer group of materials that is being 3D-printed with various levels of success. However, ceramic parts have to undergo firing and glazing processes just like any other ceramic material after designing.

A futuristic vision

An area of development with the current systems is the increase in efficiency. Theoretical efficiency with CdTe ranges at about 30 per cent, which, in commercial implementation, drops to 16 to 20 per cent.

Perovskites have been very efficient and fast developing. Higher efficiency from these materials would eventually allow you to go for reasonable-sized solar panels. One day, maybe, your smartwatch would be made of OLED for display, which could be 3D-printed and solar-charged. Yeah, wishful thinking does make the future look nice!



Fig. 3: Perovskite solar cell structure

Materials with increased efficiencies could very well usher in an era of dependence on renewable sources of energy for solar cells or OLEDs for lighting and display.

e-Textiles is an area of promise with OLEDs and solar cells. You never know, two years down the line a smartphone might be obsolete and we could have an advanced suit that can process and analyse based on the data around us. Certainly, a big help for the differently-abled! **EFY**

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